

UNIT 1

Equipment, Materials and Workplace Environment

Ampliación Lengua Extranjera Profesional Inglés



Course Evaluation

Activities & Attitude
25%

First milestone
Podcast 25%

Second milestone
Brand Fair 25%

Final Exam
25%

⚠ To pass, it is mandatory to achieve at least 5 in both milestones and the final test, as the Learning Outcomes (LOs) are assessed in all of them. If they are not passed, they will have to be retaken in the ordinary sitting and, ultimately, in the extraordinary sitting. The second milestone, being group-based, will be adapted so that it can be completed and assessed individually in those sittings.



Attendance and Continuous Assessment Rules

 You need a 7 or higher in English subjects to study in York, access Erasmus or work in the U.S.

Loss of 15% in a term

If the student exceeds 15% absences in a term, they must sit the checkpoints or tasks completed during that period in the ordinary exam session, even if they had previously passed some of them.

Loss of 20% in the full course

If the absence rate exceeds 20% over the entire course, the student must sit all tasks in the ordinary exam session, even if they have already passed some of them.

If the student do not pass in the extraordinary session

The student would have to re-enrol in a later course in order to be assessed again.



Contents of Unit 1

This unit is organised into five major content areas: vocabulary, reading, writing, listening, and speaking, always applied to formal and professional contexts. Namely:

Vocabulary

Workplace zones, equipment, materials, and key language for purpose and safe handling

Reading

Reading comprehension texts related to applied workplace English

Listening

Listening activities related to applied workplace English

Speaking

Tool/Asset Demo

Writing

Comparison Writing



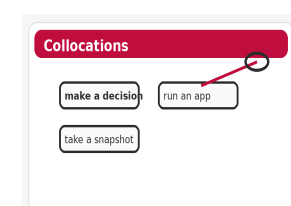
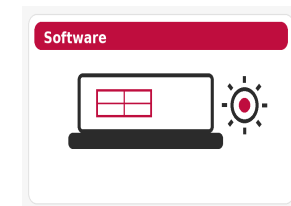
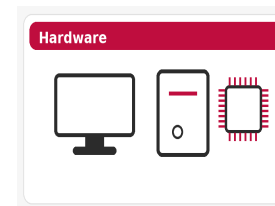
Vocabulary

In this unit we focus on the technical environment you work with every day: equipment, materials and vocabulary related to the workplace.

We will explore the hardware that underpins infrastructure: servers and their internal components, networking gear, and power and rack equipment. You will learn to describe what each component is used for and how to present its specifications clearly and professionally.

Next, we cover the software layers that make systems and applications run: operating systems and drivers, virtualization and containers, build and deployment tooling, and monitoring. We will practice concise process descriptions for installation and maintenance tasks.

Finally, we look at essential elements of physical security in technical spaces because protecting the environment is part of technical communication too.



Grammar pill: What is a collocation?

A collocation is a group of words that naturally go together. Learning these combinations helps you sound more fluent and natural in English.

For example, we say:

make a decision — not do a decision

take a snapshot — not make a snapshot

run an application — not do an application

We will be working with collocations in the following slides



Hardware

This section introduces core computer hardware used in technical environments. You will learn the names, functions, and collocations associated with key components and how to describe them professionally.



Hardware vocabulary and collocations

Term	Example sentence	Common collocations	Term	Example sentence	Common collocations
motherboard	Before upgrading the processor, check whether it is compatible with the motherboard.	replace a motherboard; motherboard compatibility	solid-state drive / SSD	Installing an SSD significantly reduced the time the computer took to start up.	install an SSD; upgrade to an SSD; SSD capacity
processor / CPU	A powerful processor enables users to run demanding applications efficiently.	a powerful processor; upgrade the processor; processor speed	power supply unit / PSU	The new graphics card requires a higher-wattage power supply unit.	replace the power supply unit; a faulty power supply unit; PSU wattage
graphics card	The design team needs a dedicated graphics card to edit high-resolution videos smoothly.	a dedicated graphics card; install a graphics card; upgrade the graphics card	heat sink	The heat sink helps transfer heat away from the processor, preventing it from overheating.	install a heat sink; a processor heat sink; heat sink fins
memory / RAM	Upgrading the RAM would allow employees to switch between applications more smoothly.	upgrade the RAM; install additional RAM; insufficient memory	expansion slot	The motherboard has a spare expansion slot for an additional network card.	an available expansion slot; occupy an expansion slot; insert a card into an expansion slot
hard disk drive / HDD	The company replaced the ageing hard disk drive after several files became inaccessible.	replace a hard drive; hard drive failure; hard drive capacity	network interface card / NIC	The technician installed a new network interface card to provide a faster wired connection to the company network.	install a network interface card; replace a faulty NIC; configure a network interface card



Let's practice: Hardware

Read the workplace situations below. Choose the correct component from the box to complete each sentence. Use each term only once.

motherboard / processor / graphics card / memory (RAM) / hard disk drive (HDD) / solid-state drive (SSD) / power supply unit (PSU) / heat sink / expansion slot / network interface card - NIC

1. "We need to connect several internal components to the computer's main circuit board. Unfortunately, this board is damaged and must be replaced." We need a new _____.
2. "Our computers slow down when we keep several applications open. We need more temporary working space for the programs currently in use." We should install additional _____.
3. "The design team needs a dedicated component to process images and handle high-resolution video more efficiently." We should upgrade the _____.
4. "We need a storage device with no moving parts that will help the computer start up and load applications quickly." We should install a _____.
5. "The technician needs to fit an additional card into the motherboard, but there are no suitable connectors available." The motherboard needs an available _____.



Let's practice: Hardware (continued)

motherboard / processor / graphics card / memory (RAM) / hard disk drive (HDD) / solid-state drive (SSD) / power supply unit (PSU) / heat sink / expansion slot / network interface card - NIC

6. "This component carries out instructions and performs the calculations needed to run programs. We need a faster one for demanding tasks."
We should upgrade the _____.
7. "Our new components require more electrical power than the existing unit can provide."
We need to replace the _____.
8. "This metal component transfers heat away from the processor. Dust has accumulated between its fins."
The technician needs to clean the _____.
9. "The computer needs an internal adapter that will allow it to connect to the company's wired network."
We should install a _____.
10. "This storage device uses spinning magnetic disks. It has started making unusual clicking noises, and some files are no longer accessible."
The technician suspects a faulty _____.



Software

This section will cover the software that makes your equipment work: the operating system, virtualisation and containers, deployment environments and tools, and observability. You will learn how to describe what each layer is used for, state basic specs/compatibility, and outline simple install/update workflows. You will also learn common operating systems used at work.



1. What is an operating system, and how is it different from an application? Give examples of both and explain what an operating system allows you to do.
2. Which operating systems have you used? Compare two of them and explain which you would recommend for studying or working, giving reasons for your choice.



Operating systems in the workplace

The Operating System (OS) is the software that manages hardware and provides services for programs.

Learn the following OS used in the workplace:

Windows — Widely used desktop OS with broad driver and software support.

Linux (distribution) — Family of OS variants (e.g., Ubuntu, Fedora) with package managers for installing and updating software.

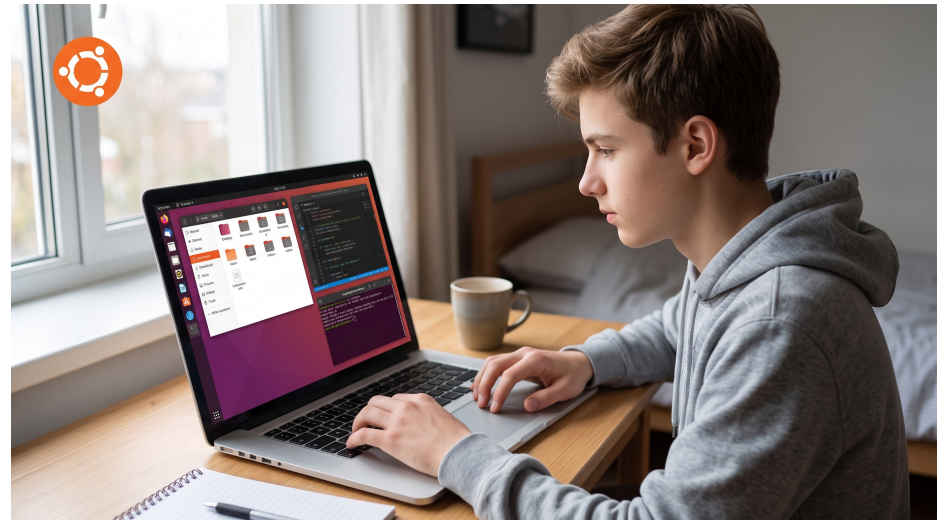
macOS — Apple's desktop OS integrated with Apple hardware and services.

iOS / Android — Mobile operating systems for phones/tablets managed in enterprise (MDM, app policies).



Vocabulary - Software

Below you will review the main software components and technical tools commonly used in professional IT environments, together with example uses and collocations.



Software components and tools

Term	Definition	Example
driver	Program that lets the OS control a hardware device.	"I need to install a driver for my printer."
kernel	The core component of an OS that manages resources and hardware–software communication.	"The new kernel improves scheduling."
service / daemon	A background program that provides functions for other software.	"Restart the web service."
package manager	Tools that install, update, and manage software packages.	"Update the package with the manager."
dependency	A software component required by another program to run.	"Install all dependencies before building."

Term	Definition	Example
hypervisor	Software that enables multiple virtual machines on a single physical machine.	"The hypervisor runs three VMs."
virtual machine (VM)	A software-based computer that runs an OS like a physical machine.	"Provision a VM for testing."
image (system/software)	A packaged copy of software or a system used for deployment.	A packaged copy of software or a system used for deployment.
snapshot	A saved state of a system or VM at a specific time.	"Take a snapshot before updates."
container	A lightweight, isolated environment for running applications.	"Run the app in a container."
orchestrator	A platform that automates deployment and management of containers.	"The orchestrator scales services."



Let's practice: Spot and correct the mistake

Each sentence contains ONE incorrectly used term in bold. Replace it with the correct term from the box. Then explain your correction in your own words. Use each term once.

driver / kernel / service -daemon / package manager / dependency / hypervisor / virtual machine (VM) / image / snapshot / container / orchestrator

1. Before updating the **virtual machine**, take a driver so that you can restore its previous state if something goes wrong.
2. The printer is not working because its **orchestrator** is missing. Install it so that the operating system can communicate with the device.
3. The developer used a **snapshot** to install, update and remove software packages on her Linux computer.
4. The application cannot run because a **hypervisor** is missing: it requires a software library that has not been installed.
5. The company uses a **dependency** to run several virtual machines on a single physical server.



Let's practice: Spot and correct the mistake (continued)

driver / kernel / service -daemon / package manager / dependency / hypervisor / virtual machine (VM) / image / snapshot / container / orchestrator

6. The technician created a **package manager** with its own operating system to test unfamiliar software without changing the host computer's setup.
7. The IT department prepared a standard service containing the **operating system** and applications for deployment on new laptops.
8. The application runs inside a lightweight **kernel**, which isolates it from other applications while sharing the host operating system's kernel.
9. As demand increased, the **image** automatically deployed additional containers to handle the extra workload.
10. A **virtual machine** runs in the background, providing printing functions to other programs. Restarting this background program cleared the print queue problem.
11. The **container** is the central component of the operating system, responsible for managing resources and communication between software and hardware.



Physical security



Physical security protects workplace assets and service continuity. Finally, we will go over access control and surveillance, redundancy and recovery, and basic server-room/workstation practices. Use clear workflow markers to describe procedures.



Physical security and continuity vocabulary

Term	Definition	Example
access control	Mechanisms to manage who can view or use resources in a computing environment.	"Access control restricts entry."
badge	A card used to authenticate and allow entry to secure areas.	"Scan your badge to enter."
CCTV	A video system for surveillance and security monitoring.	"Review CCTV footage after the incident."
secure area	A space with controlled access to protect equipment and data.	"Only staff can enter the secure area."
audit log	A record used to track actions for monitoring and compliance.	"Add the change to the audit log."

Term	Definition	Example
failover	Switching to a redundant system or network when a failure occurs.	"Test failover monthly."
redundancy	Having duplicate systems to ensure availability.	"Add redundancy to avoid downtime."
backup	A copy of data stored separately for recovery.	"Run a daily backup at midnight."
restore	The process of returning data or systems from a backup.	"Restore the files after the failure."



Let's practice: Physical security and continuity

Complete each gap with one word.

badge, visitor log, CCTV, redundancy, failover, backup, restore, restrict, throughput

⚠ There is one extra term you won't need to use.

Entry to the server room requires a _____ and all visits are written to the _____. Inside, cameras provide _____ to deter misuse. To ensure availability, core systems use _____: if the primary fails, traffic moves during _____. We schedule a nightly _____ and test a periodic _____ to verify that data can be recovered. Finally, we _____ access for non-staff and lock the rack after maintenance.



Pre-reading

Critical Operating System Flaw Raises Concerns Across the IT Industry

Discuss the following questions with a partner.

1. What do you think has happened?
2. Who is likely to be affected by this issue?
3. What topics do you expect the article to cover?
4. Do you think this is a technical problem, a cybersecurity problem, or both? Explain.
4. Predict three words or expressions that you think will appear in the article.



Reading: Pre-Reading Vocabulary Inference

Below are some words taken from the article. Without using a dictionary, work with a partner and complete the table. Place each word in the most appropriate category.

vulnerability - exploit - security patch - legacy - infrastructure - system administrator - maintenance - permissions - cybercriminal - malware - device - network - breach- update - threat

Security	Software	Hardware	People	Actions



Reading: Critical Operating System Flaw Raises Concerns Across the IT Industry

London, 9 July 2026 — Technology companies and cybersecurity agencies have issued an urgent warning following the discovery of a critical vulnerability affecting several widely used operating systems. Although software developers have already released security updates, experts fear that thousands of organisations will remain exposed because many users postpone or ignore system updates.

The vulnerability, which allows attackers to gain unauthorised access to devices, has reignited the debate over the importance of maintaining operating systems throughout their entire life cycle. While many people tend to focus on installing antivirus software or choosing strong passwords, cybersecurity professionals argue that keeping the operating system fully updated is equally essential.

According to researchers, the problem is not simply the existence of security flaws. Every complex operating system contains vulnerabilities that are discovered over time. The real challenge lies in how quickly organisations identify these weaknesses, distribute security patches, and ensure that every device receives them before cybercriminals exploit them.



Reading text (continued)

Many businesses still depend on legacy operating systems because replacing them can be both expensive and technically challenging. Older industrial machines, medical equipment and financial systems often rely on software that is incompatible with modern operating systems. As a result, companies sometimes face a difficult decision: continue using outdated systems that increase security risks or invest significant resources in upgrading their infrastructure.

Experts also point out that operating systems are no longer limited to desktop computers. Smartphones, tablets, servers, smart televisions and even household appliances rely on operating systems to manage hardware resources and execute applications. Consequently, a single security vulnerability can potentially affect millions of interconnected devices around the world.

The incident has also highlighted the growing importance of IT professionals who specialise in system administration. Their responsibilities extend far beyond installing software; they must monitor network activity, manage user permissions, implement backup strategies, and respond rapidly whenever new vulnerabilities are disclosed.



Reading text (continued)

Cybersecurity analysts believe that technical knowledge alone is no longer sufficient. They argue that organisations should promote a culture of digital responsibility in which every employee understands the consequences of neglecting routine maintenance. A delayed software update may seem insignificant, but in many cases it is precisely the opportunity that cybercriminals are waiting for.

As digital transformation accelerates across every sector of society, experts agree on one point: operating systems are not simply software hidden behind a computer screen. They represent the foundation upon which modern digital life depends, making their security, stability and continuous maintenance more important than ever before.



Let's practice: Full answer questions

Read the text related to critical operation systems and answer the following comprehension questions using full sentences.

1. Because all the operating systems have or will have vulnerabilities. The problem can be fixed fast by the technicians but if the user don't update his device, the version of his OS have still the vulnerability.

1. Why do cybersecurity experts believe that updating an operating system is just as important as installing antivirus software?

2. According to the article, what is the real challenge regarding software vulnerabilities? Explain it in your own words.

2. The real challenge it's identify and fix the vulnerabilities discovered as quick as possible, distribute security patches and ensure that the users update them OS versions before can be attacked.

3. Why do some organisations continue using legacy operating systems despite the security risks involved?

3. Because change all devices OS it's a expensive operation and some organisations prefers to run the risk instead of invest a lot of money on that.

4. The article mentions that operating systems are used in more devices than just computers. Why does this make cybersecurity more complex?

4. Because actually there are more devices that needs an OS to work and a single vulnerability can affect millions of interconnected devices around the world.



Further discussion questions

5. Explain the meaning of the sentence: "A delayed software update may seem insignificant, but in many cases it is precisely the opportunity that cybercriminals are waiting for."

It means that a delay in the distribution of an OS update that have a security patch it's the perfect opportunity for cybercriminals to exploit a vulnerability along more time.

6. In your opinion, which argument presented in the article is the most convincing? Support your answer with evidence from the text.

7. What consequences could a company face if it decides not to update its operating systems regularly?

They can have serious problems with them systems, confidential, users or clients data, that affects directly their reputation or worst have a legal problem. Because not update them can provoke a security breach tha can be exploted.

8. Do you agree that every employee—not only IT professionals—should be responsible for cybersecurity? Explain your answer using ideas from the article and your own opinion.

I totally agree, because cibersecurity not only depends on the system, it also depends on how the user use the services, all of us needs to be careful with weak passwords for example or with maintain the system updated, to prevent security breaches.



Let's practice: Vocabulary in context - Synonyms

1. Find a word or expression in the text that means to take advantage of a weakness. exploit

2. What does the word legacy mean in the expression legacy operating systems? An old version of the OS, it's outdated because it haven't support.

3. In the context of the article, what is a security patch? it's the fix implementation to a vulnerability

4. What does the verb disclose mean in the sentence "whenever new vulnerabilities are disclosed"?
It's like after been knowing

5. Explain the meaning of the expression digital transformation as it is used in the article.



Listening: True or False? (Conversation 1)

Listen to the first conversation and decide whether the following statements are true (T) or false (F). If False, justify your answer with your own words.

1. They verified grounding and power circuits before mounting the server.
2. They skip cable labelling to save time.
3. SFP modules are pre-matched to the switch.
4. They prefer updating the kernel before taking a VM snapshot.
5. They expect NVMe to improve p95 latency versus HDD.
6. Throughput is ignored on the monitoring dashboard.
7. Badge access is restricted during maintenance.
8. CCTV does not cover the patch panel.



Listening: True or False? (Conversation 2)

Listen to the second conversation and decide whether the following statements are true (T) or false (F). If False, justify your answer with your own words.

1. The pod restart is linked to a missing environment variable.
2. They made a database schema change in this release.
3. They will watch p95 latency and CPU during the canary.
4. The canary immediately routes 100% of traffic.
5. Rolling back depends partly on error budget burn.
6. They avoid VM snapshots to keep the pipeline fast.
7. The rollout note compares version 1.4.2 to 1.4.1.
8. Secrets and permissions are left unchanged for speed.



Listening: True or False? (Conversation 3)

Listen to the third conversation and decide whether the following statements are true (T) or false (F). If False, justify your answer with your own words.

1. Badge scans are optional if CCTV is working.
2. CCTV now includes the patch panel.
3. The failover test aims to confirm the standby becomes active.
4. They don't measure latency or packet loss during switchover.
5. The restore drill validates the Recovery Time Objective.
6. Mislabeled cables from last week are ignored to save time.
7. They plan to add photos and an incident timeline to the audit log.
8. Ad-hoc changes are allowed during the drill if someone approves verbally.



Speaking – Tool Demo



Prepare and present a short demo of ONE item used in your professional area (hardware, software component, or workplace element). Explain it clearly as if you were onboarding a new colleague.

Timing: 60–90 seconds, supported by one photo/image or a single slide.

Focus: clear purpose, concrete specs, risks/precautions, comparison, and one best practice.



Speaking – Demo outline

Choose **ONE** item from Unit 1 and draft a five-line outline:

Purpose: what it is and what it's for (e.g., "This UPS is used to protect servers from power loss.")

3–4 key features/specs (numbers help: capacity, ports, dimensions, power).

One potential risk + one safety precaution (e.g., "Overheating; you should keep clear airflow.")

One brief comparison ("Compared to X, this model/option...")

One best practice from real operations (e.g., "Label cables by rack unit and device.")

 **Required duration: around 2 minutes**



Tips to prepare your Demo

Keep it simple: one idea per sentence; avoid long lists.

Use numbers clearly: “Capacity up to 3000 VA”, “2 × 10GbE ports”, “M.2 2280 form factor”.

Signal your structure with signposting:

Purpose: “It is used to...”

Features: “Key features are...”

Risk/precaution: “A common risk is... so you should...”

Comparison: “Compared to..., this option is...”

Tip: “In practice, it’s best to...”

Time yourself before presenting.



Useful language

Describe purpose

“It is used to host virtual machines/containers.”

“It helps to reduce latency / keep services available.”

Give specifications

“Rated capacity: 3000 VA / 900 W.”

“Ports: 2 × 10GbE SFP+, 4 × 1GbE RJ-45.”

“Compatible with Linux kernel 6.x / supports VLAN tagging.”



Useful language (continued)

Name components/materials

“This unit includes hot-swap PSUs and front-to-back fans.”

“The image comes with pinned dependencies and a health check.”

Risks and precautions

“A common risk is overheating; you should keep 10 cm of clearance and monitor fan alarms.”

“You must not exceed the UPS power rating; check PSU draw per circuit.”

“Secure the rack and restrict access; badge entry only.”



Useful language (comparison and best practice)

Compare options

“Compared to HDD, NVMe offers lower latency and higher throughput.”

“Compared to VMs, containers are more lightweight and start faster.”

Add a best practice

“Always take a VM snapshot before major updates.”

“Label every cable by rack unit and device; attach a photo to the audit log.”



Oral presentation checklist

Make sure you complete this checklist before the oral presentation:

- ☐ I have 1 clear purpose sentence.
- ☐ I have 3 concrete specs with numbers.
- ☐ I mention 1 risk + 1 precaution.
- ☐ I include 1 short comparison.
- ☐ I close with 1 best practice/tip.
- ☐ Last but not least: I include specific vocabulary that I learned in class and my speech is clear.



Writing: Comparison Writing

Write a short, professional comparison (120–150 words) between two IT items from Unit 1. Keep the tone concise and technical, focusing on equipment, materials, or workplace environment. Explain purpose, key specs, similarities, differences, and end with a reasoned recommendation.

Choose ONE pair of items:

Laptop vs. Desktop PC

SSD vs. HDD

Wi-Fi (access point) vs. Ethernet cable

Cloud storage vs. External hard drive

Linux vs. Windows



Writing requirements

What you must include:

- Introduction: context and purpose of the comparison (e.g., lab upgrade, office rollout, server refresh).
- At least three similarities (what both items do/share: purpose, basic function, environment).
- At least three differences (specs, capacity, latency, throughput, power/space, handling, risks).
- Conclusion: which option is preferable and why, given one constraint (space, budget, latency target, uptime, manageability).

Required length: 120–150 words. Register: professional and precise.



Useful connectors

Here is a set of connectors you can use to structure your writing task:

Similarity: similarly; likewise; both X and Y; in the same way

Difference: however; whereas; while; in contrast; on the other hand

Adding points: moreover; furthermore; in addition

Cause/result: therefore; thus; as a result

Framing: to begin with; overall; in conclusion; given this



Key vocabulary: Laptop vs. Desktop PC

Below, you will find key vocabulary you can use for your task. Use ONLY what fits your comparison writing:

Laptop vs. Desktop PC

portability, battery life, charger, integrated webcam, built-in keyboard/trackpad

desktop tower, external monitor, peripherals, desk space, upgradeability

performance, thermal limits, fan noise, power draw, heat, docking station

use cases: remote work, office workstation, gaming, 3D rendering



Key vocabulary: SSD vs. HDD

SSD vs. HDD

solid-state drive (SSD), hard disk drive (HDD), flash memory, spinning disk, platter

read/write speed, latency, IOPS, capacity (GB/TB), endurance

reliability, shock resistance, noise, heat

SATA, NVMe, form factor (2.5", M.2), boot time



Key vocabulary: Wi-Fi vs. Ethernet

Wi-Fi (access point) vs. Ethernet cable

wireless signal, coverage, interference, bandwidth, latency, stability

Ethernet cable, RJ-45 connector, CAT6, wired connection, throughput

access point (AP), SSID, 2.4 GHz / 5 GHz / 6 GHz, WPA3 security

packet loss, ping, jitter, line-of-sight, wall attenuation



Key vocabulary: Cloud storage vs. External hard drive

Cloud storage vs. External hard drive

cloud storage, sync, version history, sharing permissions, redundancy

external HDD/SSD, offline access, capacity, portability

bandwidth, upload/download speed, subscription cost

backup strategy, encryption, data privacy



Key vocabulary: Linux vs. Windows

Linux vs. Windows

operating system (OS), distro, package manager, command line/terminal

driver support, compatibility, software ecosystem, updates

customization, resource usage, licensing, productivity suite





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